Project Initialization and Planning Phase

Date	15 March 2024
Team ID	SWTID-2026-2085
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed **OptiCrop: Smart Agricultural Production Optimization Engine** aims to assist farmers in selecting the most suitable crop by leveraging machine learning techniques. The system analyzes soil nutrients and environmental conditions to provide accurate crop recommendations, improving agricultural productivity, reducing crop failure, and supporting data-driven farming decisions.

Project Overview	
Objective	The primary objective of this project is to develop an intelligent crop recommendation system that uses machine learning to analyze soil nutrients and environmental conditions, enabling farmers to make informed crop selection decisions and improve agricultural productivity.
Scope	The project focuses on predicting the most suitable crop based on soil nutrients (Nitrogen, Phosphorus, Potassium), pH level, temperature, humidity, and rainfall. It provides farmers with accurate recommendations through a simple web-based application.
Problem Statement	
Description	Farmers often face challenges in selecting the right crop due to changing weather conditions, varying soil quality, and limited access to expert agricultural guidance. Incorrect crop selection can result in low productivity and financial losses.

Impact	By providing accurate crop recommendations, the system helps improve crop yield, reduce farming risks, optimize resource utilization, and increase farmers' income through better decision-making.
Proposed Solution	
Approach	The system uses a Machine Learning classification model trained on agricultural datasets. After receiving soil and environmental parameters from the user, the trained model predicts the most suitable crop and displays the recommendation through a user-friendly web interface.
Key Features	• Machine Learning-based crop recommendation. • Prediction using soil nutrients and environmental data. • Fast and accurate crop prediction. • Simple and user-friendly web interface. • Reliable decision support for farmers. • Scalable architecture for future enhancements such as weather API integration and fertilizer recommendations.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 (4+ Cores) or AMD Ryzen 5/7
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	256 GB SSD (Minimum)
Software		

Frameworks	Python frameworks	Flask
Libraries	Machine Learning and Data Processing Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Pickle
Development Environment	IDE	Visual Studio Code, PyCharm, Jupyter Notebook
Data		
Data	Source, size, format	Crop Recommendation Dataset, Kaggle, CSV (≈2200 records, 8 features + crop label)